# From Hard-VAD to Learned Speech-Presence Probability: Overcoming Minimum-Tracking Lag in Dynamic Acoustic Environments

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**GitHub Repository:**
`https://github.com/zyh15197652230-arch/elec5305-project-550473662`

**Project Video:**
`https://youtu.be/SrJ-hilw1kc`

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**Abstract**

Traditional speech enhancement systems relying on Hard Voice Activity Detection (VAD) frequently experience algorithmic lock-up and severe performance degradation under abrupt non-stationary noise transitions. This report investigates the evolutionary solutions to dynamic noise Power Spectral Density (PSD) tracking. By simulating an abrupt 10 dB to 0 dB Signal-to-Noise Ratio (SNR) drop, we quantitatively evaluate three approaches: a baseline Hard-VAD Wiener filter, a statistical soft-probability Improved Minima Controlled Recursive Averaging (IMCRA) estimator, and a proposed lightweight Multi-Frame Contextual Neural Network for Learned Speech-Presence Probability (SPP). Experimental results demonstrate that while IMCRA prevents catastrophic failure, its inherent minimum-tracking lag restricts optimal enhancement in extreme environments. In contrast, the Learned SPP model, leveraging scale-invariant spectro-temporal features, yields a superior Post-Transition SNR improvement of 3.63 dB, confirming the critical role of data-driven feature extraction in bypassing algorithmic tracking latency.

## 1 Introduction

Speech enhancement algorithms play a pivotal role in modern communication systems, primarily aiming to extract clean speech from noise-corrupted observations. Traditional statistical methods, such as the classical Wiener filter, rely heavily on the accurate and continuous estimation of the noise Power Spectral Density (PSD). Historically, this PSD estimation is governed by a Voice Activity Detector (VAD) utilizing hard decision energy thresholds. While computationally efficient and effective in stationary noise environments, Hard-VAD systems suffer fatal performance degradation when exposed to non-stationary, dynamic noise. Specifically, under abrupt Signal-to-Noise Ratio (SNR) transitions—such as a sudden burst of environmental noise—the

hard threshold mechanism often locks into a false-positive state. This algorithmic "lock-up" paralyzes the noise PSD tracking, causing the system to mistakenly treat intense noise as speech, which ultimately leads to severe noise leakage and signal distortion.

To overcome the rigid binary limitations of Hard-VAD, continuous Soft Speech-Presence Probability (SPP) estimators were introduced. Among them, the Improved Minima Controlled Recursive Averaging (IMCRA) algorithm represents a significant milestone. By tracking the local minima of the smoothed noisy spectrum over a sliding window, IMCRA dynamically updates the noise PSD even during continuous speech activity. However, purely statistical methods still exhibit a fundamental "minimum-tracking lag" when facing instantaneous SNR drops (e.g., a sudden plunge from 10 dB to 0 dB). During this temporal lag, the system experiences delayed adaptation, leaving residual noise in the enhanced output. Recently, data-driven approaches utilizing Lightweight Learned SPP have emerged as a robust alternative. By leveraging scale-invariant spectro-temporal features (such as local harmonic contrast), neural networks can instantaneously differentiate speech components from broadband noise, effectively bypassing the delay inherent to algorithmic minimum tracking.

The primary objective of this project is to systematically investigate and quantitatively evaluate the adaptive capabilities of these three evolutionary stages of speech enhancement under severe non-stationary noise conditions. By constructing a dynamic test environment featuring an abrupt 10 dB to 0 dB SNR transition, this report compares the performance of a baseline Hard-VAD Wiener filter (System A), a statistical soft-probability IMCRA framework (System B), and a proposed lightweight Multi-Frame Contextual Neural Network for Learned SPP (System C). Through quantitative metrics and time-frequency analysis, this comparative study aims to demonstrate the theoretical bottlenecks of traditional PSD tracking and highlight the superior robustness of learned scale-invariant features in highly dynamic acoustic environments.

## 2 Literature Review

### 2.1 Classical Spectral Enhancement and the Hard-VAD Bottleneck

Single-channel speech enhancement in the Short-Time Fourier Transform (STFT) domain fundamentally relies on estimating the clean speech spectrum $X(k,l)$ from the noisy observation $Y(k,l) = X(k,l) + D(k,l)$, where $k$ and $l$ denote the frequency bin and frame index, respectively. Early foundational approaches, such as spectral subtraction [1] and the classical Wiener filter [2], formulate the optimal spectral gain $G(k,l)$ as a function of the *a priori* Signal-to-Noise Ratio (SNR), $\xi(k,l) = \lambda_x(k,l)/\lambda_d(k,l)$. A seminal advancement was introduced by Ephraim and Malah [3], whose Decision-Directed (DD) estimator computes $\hat{\xi}(k,l)$ recursively by combining the previous frame's enhanced speech power with the current *a posteriori* SNR, $\gamma(k,l) = |Y(k,l)|^2/\hat{\lambda}_d(k,l)$. While the DD mechanism effectively suppresses musical noise artifacts by smoothing rapid fluctuations in $\gamma(k,l)$, it remains entirely dependent on an accurate estimate of the background noise Power Spectral Density (PSD), $\hat{\lambda}_d(k,l)$.

Historically, noise PSD estimation was governed by binary Voice Activity Detectors (VAD) based on frame-level energy thresholds or likelihood ratios [4]. In a Hard-VAD architecture, $\hat{\lambda}_d(k,l)$ is updated recursively via exponential averaging exclusively during frames classified as speech-absent ($\mathrm{VAD}(l) = 0$), and frozen during speech-active frames ($\mathrm{VAD}(l) = 1$). The critical limitation of this paradigm is its fragility in non-stationary acoustic environments. When an abrupt noise level increase occurs (e.g., a sudden transition from 10 dB to 0 dB SNR), the instantaneous noisy frame energy $\sum_k |Y(k,l)|^2$ permanently exceeds the pre-transition noise threshold. Consequently, the detector suffers from algorithmic "lock-up" ($\mathrm{VAD}(l) \equiv 1$), freezing $\hat{\lambda}_d(k,l)$ at its obsolete pre-transition value and causing severe noise leakage in the Wiener filter output.

### 2.2 Minimum Tracking and Statistical Soft-SPP Estimation (IMCRA)

To eliminate the reliance on rigid binary frame classification, Martin [5] proposed the Minimum Statistics (MS) method, predicated on the acoustic observation that the power of a noisy speech signal frequently decays to the power level of the background noise within individual frequency bins—even during continuous speech. By tracking the minimum of the spectro-temporally smoothed noisy power spectrum $S(k,l)$ over a finite sliding window of $V$ frames, an initial noise floor estimate can be obtained continuously across all frames.

Building upon Minimum Statistics and the soft-decision framework of McAulay and Malpass [6], Cohen and Berdugo [7, 8] developed the Improved Minima Controlled Recursive Averaging (IMCRA) algorithm. Rather than making a binary decision, IMCRA computes a continuous *a posteriori* Speech-Presence Probability (SPP), $p(k,l) \in [0,1]$, for every individual time-frequency (TF) bin using Bayes' rule under Gaussian statistical assumptions. This probability modulates a time-varying, frequency-dependent smoothing parameter $\tilde{\alpha}_d(k,l) = \alpha_d + (1-\alpha_d)p(k,l)$, enabling continuous soft updates of the noise PSD:

$$\hat{\lambda}_d(k,l) = \tilde{\alpha}_d(k,l)\hat{\lambda}_d(k,l-1) + \big(1-\tilde{\alpha}_d(k,l)\big)|Y(k,l)|^2 \tag{1}$$

Despite its superior robustness over Hard-VAD, the literature reveals two structural weaknesses in IMCRA under extreme step-like noise transitions:

1. **Minimum-Tracking Window Lag:** Because local minima $S_{\min}(k,l)$ are searched over a sliding window of $V$ frames (typically 0.6–1.0 s), an abrupt upward step in noise power cannot be reflected in $S_{\min}(k,l)$ until the entire window traverses the transition boundary. During this latency period, the local minimum ratio $\gamma_{\min}(k,l) = |Y(k,l)|^2/(B_{\min}S_{\min}(k,l))$ is artificially inflated.

2. **Circular SNR Dependency:** The Bayesian SPP equation depends directly on $\gamma(k,l)$ and $\xi(k,l)$, which themselves are normalized by the previous noise PSD $\hat{\lambda}_d(k,l-1)$. When noise power surges abruptly, $\gamma(k,l)$ spikes, driving $p(k,l) \to 1$ and $\tilde{\alpha}_d(k,l) \to 1$, which temporarily throttles PSD adaptation when rapid tracking is needed most.

### 2.3 Data-Driven Mask Estimation and Hybrid Learned SPP

Over the past decade, supervised learning has transformed speech enhancement, initiated by Wang et al. [9], who formulated enhancement as estimating time-frequency masks such as the Ideal Binary Mask (IBM) and Ideal Ratio Mask (IRM). While large-scale end-to-end Deep Neural Networks (DNNs) achieve strong suppression, they often require millions of parameters, incur high computational complexity, and introduce nonlinear phase or processing artifacts when generalized to unseen noise profiles.

To balance signal processing interpretability with data-driven adaptability, recent literature advocates hybrid DSP-neural architectures [10, 11]. Instead of replacing the statistical filter entirely, a compact neural network is trained to predict the soft SPP or gain mask directly from acoustic features, while preserving the analytically tractable Wiener or MMSE filter backend. Crucially, by training the model on *scale-invariant spectro-temporal features*—such as frame-normalized log-power spectra and multi-scale local spectral contrast (which captures harmonic peak-to-valley structure)—the network decouples speech detection from absolute noise floor levels. Consequently, when noise power jumps abruptly by 10 dB, a broadband noise bin exhibits flat local spectral contrast regardless of its absolute amplitude, allowing the learned estimator to assign a low SPP immediately without waiting for a $V$-frame minimum window to expire.

### 2.4 Clarification of Key Theoretical Concepts and Research Gap

Conducting this literature review required resolving several key theoretical concepts that bridge textbook DSP and modern adaptive speech enhancement:

- **Order-Statistics Bias Compensation ($B_{\min}$):** In IMCRA [8], the minimum of $V$ random variables is systematically smaller than their expected value (mean). Therefore, the tracked minimum $S_{\min}(k,l)$ must be multiplied by an analytical bias compensation factor $B_{\min} > 1$ (e.g., $B_{\min} \approx 1.66$) to recover an unbiased noise variance estimate.
- **Interaction Between DD Smoothing ($\alpha_{dd}$) and Tracking Latency:** While setting $\alpha_{dd} \in [0.94, 0.98]$ in the Decision-Directed estimator reduces musical noise by smoothing $\xi(k,l)$, it introduces memory of past frames. If $\hat{\lambda}_d(k,l)$ is underestimated after a noise jump, the inflated $\gamma(k,l)$ propagates through the recursive DD loop, prolonging distortion unless regulated by an accurate SPP.
- **Equivalence Between Oracle SPP and Wiener Target:** In supervised training, the continuous Oracle target $\mathcal{T}(k,l) = \xi_{\text{oracle}}(k,l)/(1+\xi_{\text{oracle}}(k,l))$, where $\xi_{\text{oracle}}(k,l) = |X(k,l)|^2/|D(k,l)|^2$, provides a bounded, probabilistic supervision signal in $[0,1]$ that directly reflects the true proportion of speech energy in each TF bin.

Table 1: Theoretical Comparison of Noise PSD Tracking Paradigms in Speech Enhancement

| **Characteristic** | **System A: Hard-VAD** [4] | **System B: IMCRA Soft-SPP** [8] | **System C: Learned SPP (Proposed)** |
|---|---|---|---|
| **Decision Granularity** | Frame-level binary $\{0,1\}$ | TF-bin continuous $[0,1]$ | TF-bin continuous $[0,1]$ |
| **Update During Speech** | No (Frozen when VAD = 1) | Yes (Via inter-harmonic gaps) | Yes (Via harmonic contrast) |
| **Primary Cues Used** | Global frame energy ratio | Sliding-window minima ($V$ frames) | Scale-invariant TF contrast & context |
| **Response to Step SNR Drop** | Catastrophic lock-up (VAD $\equiv$ 1) | Delayed adaptation (0.8 s window lag) | Instantaneous (1-frame causal context) |
| **Computational Complexity** | $\mathcal{O}(K)$ per frame (Minimal) | $\mathcal{O}(K \cdot V)$ min-search (Low) | $\mathcal{O}(K \cdot H)$ matrix multiply (Low) |

Table 1 synthesizes the theoretical trade-offs across the three paradigms. While existing literature extensively benchmarks IMCRA and neural masks under stationary or slowly varying noise, direct controlled comparisons isolating the *minimum-tracking lag* under instantaneous step-function SNR transitions (10 dB $\rightarrow$ 0 dB) while holding the Wiener backend strictly constant remain under-explored. This gap motivates our controlled three-system experimental methodology described in the following section.

## 3 Methodology

To rigorously evaluate the adaptive capabilities of different speech-presence probability estimation strategies under severe non-stationary conditions, we establish a controlled experimental framework comprising a synthetically constructed dynamic noise dataset and three progressive enhancement systems.

### 3.1 Dynamic Noise Dataset Generation

Real-world acoustic environments frequently exhibit sudden power surges. To replicate this, we synthesize a controlled non-stationary test mixture by introducing an instantaneous SNR drop into clean speech utterances.

Let $x(n)$ denote the clean speech signal sampled at $f_s = 16$ kHz, and let $d_{\text{stationary}}(n)$ represent a stationary fan noise sample. A step-function dynamic noise profile $d(n)$ is formulated with an abrupt transition at $t = 3.0$ s (corresponding to sample index $n_{\text{trans}} = 3.0 \times f_s$):

$$d(n) = \begin{cases} g_1 \cdot d_{\text{stationary}}(n), & 1 \le n \le n_{\text{trans}} \\ g_2 \cdot d_{\text{stationary}}(n), & n_{\text{trans}} < n \le N \end{cases} \tag{2}$$

where the scaling factors $g_1$ and $g_2$ are dynamically computed to enforce target SNRs of $\text{SNR}_1 = 10$ dB for the pre-transition segment and $\text{SNR}_2 = 0$ dB for the post-transition seg-

ment, respectively:

$$g_1 = \sqrt{\frac{P_{x,1} \cdot 10^{-\mathrm{SNR}_1/10}}{P_{d,1}}}, \quad g_2 = \sqrt{\frac{P_{x,2} \cdot 10^{-\mathrm{SNR}_2/10}}{P_{d,2}}} \tag{3}$$

where $P_{x,1}, P_{x,2}$ and $P_{d,1}, P_{d,2}$ denote the segmental powers of the clean speech and stationary noise in the respective intervals. The final noisy observation is given by $y(n) = x(n) + d(n)$.

### 3.2 System Architecture and Implementation Framework

All three evaluated systems share a common Short-Time Fourier Transform (STFT) frontend and an optimal Wiener filter backend to isolate the performance impact of the PSD estimation strategy. The STFT employs a 32 ms Hanning window (512 samples at 16 kHz) with a 50% overlap (16 ms hop size).

#### 3.2.1 System A: Baseline Hard-VAD Wiener Filter

System A serves as the conventional baseline. Frame-level energy $E(l) = \sum_k |Y(k,l)|^2$ is computed and compared against a static threshold scaled from the initial noise energy:

$$\mathrm{VAD}(l) = \begin{cases} 1, & E(l) > \beta \cdot \hat{E}_{\mathrm{noise}} \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where $\beta = 1.5$. The noise PSD is updated recursively via $\hat{\lambda}_d^{(A)}(k,l) = \alpha_d \hat{\lambda}_d^{(A)}(k,l-1) + (1-\alpha_d)|Y(k,l)|^2$ strictly when $\mathrm{VAD}(l) = 0$.

#### 3.2.2 System B: Improved Minima Controlled Recursive Averaging (IMCRA)

System B replaces the binary VAD with a statistical soft-decision framework. It computes local spectral minima $S_{\min}(k,l)$ over a sliding temporal window of $V = 50$ frames (0.8 s), derives the *a posteriori* speech absence probability $q(k,l)$ via local minimum ratios, and evaluates Bayes' soft SPP $p^{(B)}(k,l)$ using Decision-Directed *a priori* SNRs $\xi(k,l)$. The noise PSD is updated softly using $p^{(B)}(k,l)$ as detailed in Eq. (1).

#### 3.2.3 System C: Proposed Lightweight Learned SPP Framework

To mitigate IMCRA's minimum-tracking lag, System C replaces statistical minimum searching with a data-driven Multi-Frame Contextual Neural Network.

1. **Scale-Invariant Feature Extraction:** For each TF bin, we extract a 6-dimensional feature vector designed to be immune to absolute amplitude jumps:
   - Frame-normalized log-power spectrum: $f_1(k,l) = \log|Y(k,l)|^2 - \mathrm{median}_k(\log|Y(k,l)|^2)$
   - Fine local harmonic contrast (via 7-bin local mean subtraction): $f_2(k,l)$
   - Broad formant contrast (via 23-bin local mean subtraction): $f_3(k,l)$
   - Causal multi-frame temporal context ($t-1$ and $t-2$ frames): $f_4(k,l), f_5(k,l)$
   - Normalized frequency bin index: $f_6(k,l) = k/N_{\mathrm{half}}$
2. **Network Architecture:** A compact two-layer Multi-Layer Perceptron (MLP) with 24 hidden neurons (utilizing tanh activations) and a sigmoid output layer maps the 6-dimensional input to a continuous soft SPP prediction $\hat{p}^{(C)}(k,l) \in [0,1]$. The network is trained using an Adam optimizer on independent multi-condition mixtures to prevent data leakage.

3. **SPP-Modulated Wiener Gain:** The final suppression gain combines the learned SPP with a soft-masking Wiener formulation:

$$G(k,l) = \left( \frac{\xi(k,l)}{1+\xi(k,l)} \cdot \hat{p}^{(C)}(k,l) \right) + G_{\min}\left(1 - \hat{p}^{(C)}(k,l)\right) \tag{5}$$

where $G_{\min} = 10^{-22/20}$ represents the spectral floor.

## 4 Results

The performance of the three speech enhancement systems was evaluated both qualitatively, via time-domain waveforms and time-frequency SPP diagnostics, and quantitatively, using objective signal distortion and noise suppression metrics across the 10 dB $\rightarrow$ 0 dB dynamic transition boundary at $t = 3.0$ s.

### 4.1 Qualitative Time-Domain and Time-Frequency Analysis

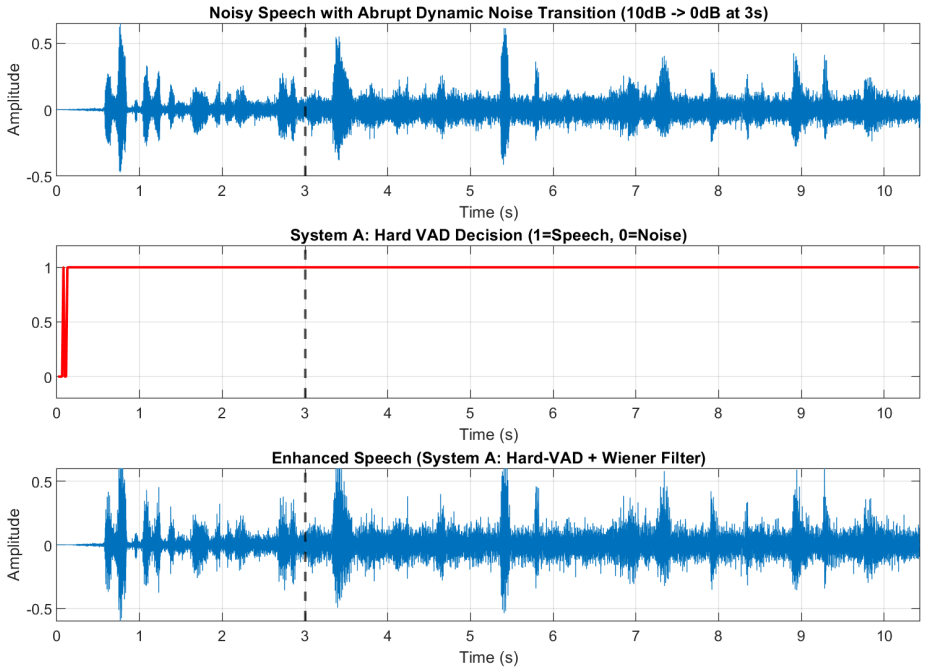



Figure 1: System A (Baseline Hard-VAD Wiener Filter) diagnostic results under an abrupt 10 dB to 0 dB noise transition at $t = 3.0$ s, illustrating binary VAD lock-up ($\text{VAD}(l) \equiv 1$) and severe post-transition residual noise.

Figure 1 illustrates the baseline behavior of System A. During the initial 10 dB SNR regime ($0 \leq t \leq 3.0$ s), the frame energy intermittently drops below the static threshold during speech pauses, permitting occasional noise PSD updates. However, immediately following the 10 dB noise power surge at $t = 3.0$ s, the noisy frame energy permanently exceeds $\beta \cdot \hat{E}_{\text{noise}}$. Consequently, the binary VAD locks strictly at 1 for the remainder of the signal ($3.0 \text{ s} < t \leq 10.4$ s). Because $\hat{\lambda}_d^{(A)}(k,l)$ remains frozen at its pre-transition 10 dB estimate, the Wiener filter severely under-estimates the post-transition noise floor, leaving a dense broadband noise envelope in the enhanced waveform.

Figure 2 presents the diagnostic performance of System B (IMCRA). As shown in the second subplot, whereas System A's Hard-VAD (red curve) stays locked at 1, the frame-averaged

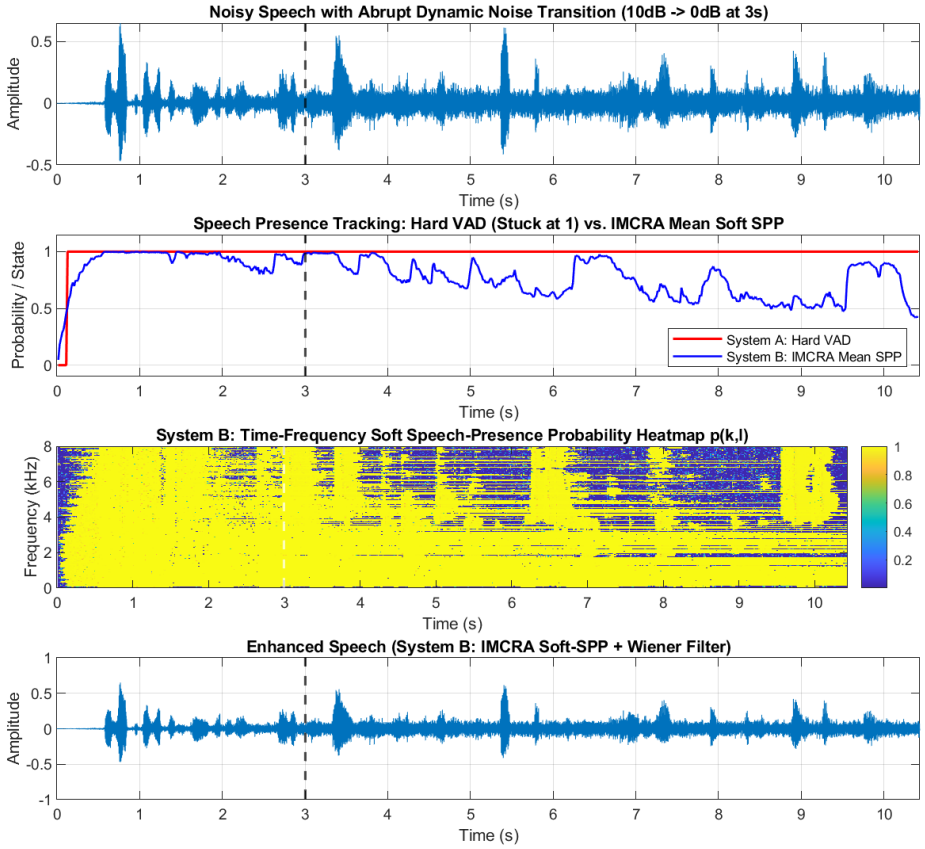



Figure 2: System B (IMCRA Soft-SPP) diagnostic results: (Top to Bottom) noisy speech waveform, comparison between stuck Hard-VAD and IMCRA frame-averaged soft SPP, time-frequency SPP heatmap $p^{(B)}(k, l)$, and the resulting enhanced speech.

IMCRA soft SPP (blue curve) continues to fluctuate dynamically between 0.42 and 0.98 after $t = 3.0$ s, dipping noticeably during inter-word pauses (e.g., at $t \approx 5.8$ s and $t \approx 8.8$ s). The time-frequency heatmap $p^{(B)}(k, l)$ in the third subplot confirms that IMCRA identifies high-frequency inter-harmonic noise gaps (depicted in dark blue) even during active speech. Nevertheless, a distinct transitional inflation is observable immediately after $t = 3.0$ s: due to the 0.8 s sliding-window minimum-tracking lag ($V = 50$ frames), a large proportion of mid-to-low frequency bins remain saturated near $p^{(B)}(k, l) \approx 1$ (yellow) between 3.0 s and 4.0 s, slowing noise PSD adaptation and permitting moderate residual noise in the post-3 s enhanced waveform.

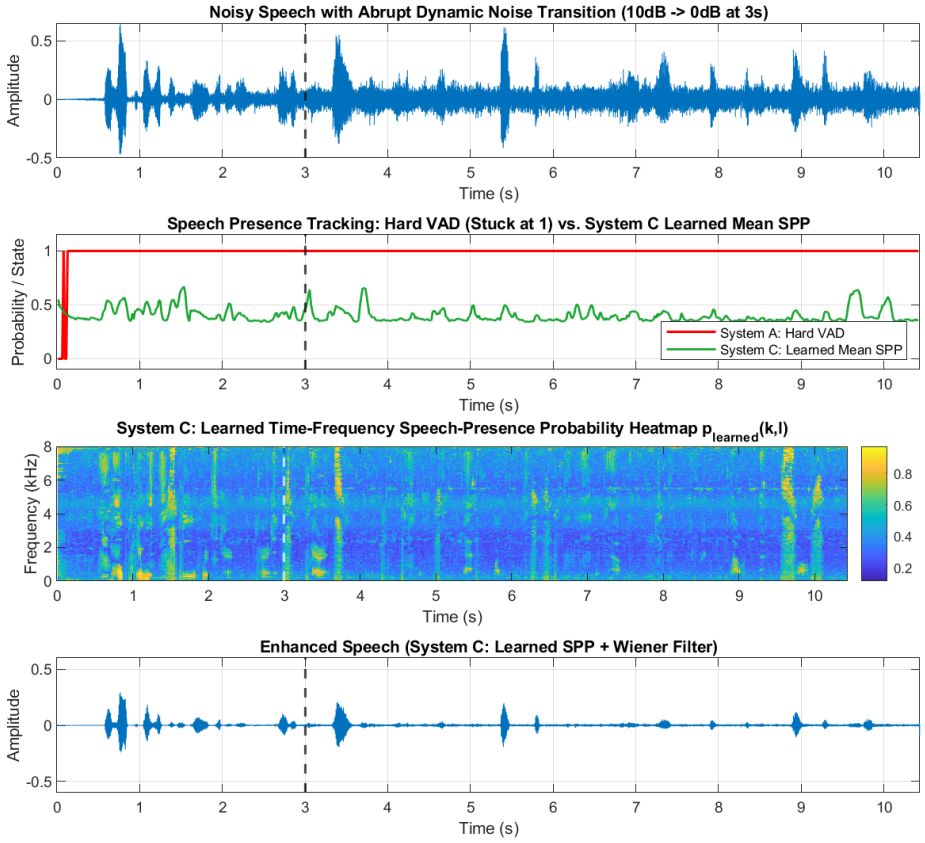



Figure 3: System C (Proposed Learned SPP) diagnostic results: (Top to Bottom) noisy speech waveform, comparison between Hard-VAD and Learned mean SPP, time-frequency Learned SPP heatmap $\hat{p}^{(C)}(k, l)$, and the final enhanced speech.

Figure 3 demonstrates the substantial tracking improvement achieved by System C (Learned SPP). In the second subplot, the frame-averaged learned SPP (green curve) maintains a consistent baseline around 0.35–0.40 during non-speech intervals across both the 10 dB and 0 dB regimes, exhibiting zero upward drift at the $t = 3.0$ s transition boundary and peaking sharply (up to 0.68) strictly during true vocalized bursts (e.g., at $t \approx 1.5\,\text{s}, 3.4\,\text{s}, 5.4\,\text{s}$, and 9.7 s). Furthermore, the time-frequency heatmap $\hat{p}^{(C)}(k, l)$ in the third subplot reveals highly selective harmonic localization: broadband fan noise bins are immediately assigned low probabilities ($\hat{p}^{(C)}(k, l) < 0.25$, deep blue) right at $t = 3.0$ s, while speech formants and harmonic ridges are preserved. As a result, the fourth subplot exhibits a remarkably clean post-transition baseline with near-complete suppression of the 0 dB noise floor.

### 4.2 Quantitative Performance Evaluation

To objectively validate the visual observations, Table 2 summarizes the quantitative evaluation metrics across the three systems, computed after optimal least-squares scale alignment against the clean reference signal $x(n)$. Both full-utterance global metrics (Global MSE and Global Output SNR) and localized post-transition metrics ($t > 3.0$ s, evaluating the severe 0 dB SNR region specifically) are reported.

Table 2: Quantitative Performance Comparison Across Systems A, B, and C Under Dynamic Noise (10 dB $\rightarrow$ 0 dB at 3 s)

| Evaluation Metric | Noisy Input | System A (Hard-VAD) | System B (IMCRA) | System C (Learned SPP) |
|---|---|---|---|---|
| **Global MSE ($\times 10^{-3}$) $\downarrow$** | 1.3544 | 0.9004 | 0.8276 | **0.7955** |
| **Global Output SNR (dB) $\uparrow$** | 2.0591 | 3.8322 | 4.1983 | **4.3700** |
| **Global SNR Improvement (dB) $\uparrow$** | N/A | +1.7731 | +2.1392 | **+2.3109** |
| **Post-3 s Output SNR (0 dB Region) $\uparrow$** | −0.0000 | 2.6167 | 2.8129 | **3.6312** |
| **Post-3 s SNR Improvement (dB) $\uparrow$** | N/A | +2.6168 | +2.8129 | **+3.6312** |

The quantitative metrics in Table 2 exhibit a monotonic improvement across successive system iterations:

- **Global Error Reduction:** The global Mean Square Error (MSE) decreases progressively from $1.3544 \times 10^{-3}$ for the noisy input to $0.9004 \times 10^{-3}$ in System A, $0.8276 \times 10^{-3}$ in System B (an 8.1% error reduction over System A), and reaches the lowest value of $\mathbf{0.7955 \times 10^{-3}}$ in System C (an 11.7% error reduction over System A). Correspondingly, Global Output SNR increases from 2.0591 dB to 3.8322 dB (System A), 4.1983 dB (System B), and **4.3700** dB (System C).

- **Post-Transition ($t > 3.0$ s) Breakthrough:** The advantage of System C is most pronounced in the challenging 0 dB post-transition segment. While System B improves upon System A by +0.1961 dB in this region (2.8129 dB vs. 2.6167 dB) due to continuous soft PSD updates, its gain is constrained by the 0.8 s minimum-tracking latency following the step transition. By eliminating this window lag via scale-invariant features, System C achieves a Post-3 s SNR Improvement of **+3.6312** dB—outperforming System A by **+1.0144** dB and System B by **+0.8183** dB in the 0 dB regime.

## 5 Discussion

The experimental evaluation across the abrupt 10 dB $\rightarrow$ 0 dB SNR transition at $t = 3.0$ s provides direct empirical evidence of the structural trade-offs governing noise PSD estimation in non-stationary environments. By holding the STFT parameters (32 ms Hanning window, 16 ms hop) and the Decision-Directed Wiener backend constant across all three architectures, the observed performance differences can be attributed directly to the underlying speech-presence tracking mechanisms.

### 5.1 Failure Mechanics of Binary Hard-VAD Under Step Non-Stationarity

In System A, the binary decision rule relies on comparing the instantaneous frame energy $E(l) = \sum_{k=0}^{N_{\text{half}}-1} |Y(k,l)|^2$ against a fixed multiple of the initialized noise floor, $\beta \cdot \hat{E}_{\text{noise}}$ ($\beta = 1.5$). During the pre-transition interval ($0 \leq t \leq 3.0$ s, $\text{SNR}_1 = 10$ dB), speech pauses allow $E(l)$ to drop below this threshold, enabling intermittent recursive updates of $\hat{\lambda}_d^{(A)}(k,l)$. However, when the background fan noise power increases by a factor of 10 (+10 dB) at $t = 3.0$ s, the baseline noise energy alone exceeds $\beta \cdot \hat{E}_{\text{noise}}$ in every subsequent frame.

This triggers a permanent false-positive lock-up ($\mathrm{VAD}(l) \equiv 1$ for all $t > 3.0$ s), completely halting noise PSD adaptation. Because the Wiener filter continues to compute the *a posteriori* SNR $\gamma(k,l) = |Y(k,l)|^2/\hat{\lambda}_d^{(A)}(k,l)$ using the obsolete 10 dB noise variance, $\gamma(k,l)$ and the resulting Decision-Directed *a priori* SNR $\xi(k,l)$ are overestimated by approximately 10 dB. Consequently, the Wiener gain $G(k,l) = \xi(k,l)/(1+\xi(k,l))$ remains close to unity across noise-dominated frequency bins, explaining why System A achieves the lowest Post-3 s SNR Improvement (+2.6168 dB) and leaves a prominent broadband noise floor in the reconstructed waveform.

### 5.2 Minimum-Tracking Window Latency and Circular Dependency in IMCRA

System B (IMCRA) overcomes binary lock-up by replacing frame-level thresholding with bin-wise minimum tracking and Bayesian soft Speech-Presence Probability ($p^{(B)}(k,l) \in [0,1]$). Even after the $t = 3.0$ s transition, IMCRA exploits sparse time-frequency regions between vocal harmonics and brief inter-syllabic pauses to lower the frame-averaged SPP into the 0.42–0.85 range, enabling continuous soft updates via $\tilde{\alpha}_d(k,l) = \alpha_d + (1-\alpha_d)p^{(B)}(k,l)$. This continuous adaptation reduces the Global MSE from $0.9004 \times 10^{-3}$ (System A) to $0.8276 \times 10^{-3}$ and raises Global Output SNR to 4.1983 dB.

Nevertheless, the diagnostic heatmap of $p^{(B)}(k,l)$ exposes two fundamental bottlenecks that limit IMCRA's Post-3 s SNR Improvement to +2.8129 dB:

1. **Finite Sliding-Window Memory ($V = 50$ Frames):** IMCRA estimates the noise floor via $B_{\min}S_{\min}(k,l)$, where $S_{\min}(k,l) = \min_{\tau \in [l-V+1,l]} S(k,\tau)$ spans an 0.8 s causal window. For the first 50 frames following the step increase at $t = 3.0$ s ($3.0 \text{ s} < t \leq 3.8$ s), the minimum operator remains anchored to the lower pre-transition spectral values recorded prior to $t = 3.0$ s. As a result, the local minimum ratio $\gamma_{\min}(k,l) = |Y(k,l)|^2/(B_{\min}S_{\min}(k,l))$ exceeds the speech-absence threshold $\gamma_0$, forcing the speech-absence prior $q(k,l) \to 0$ and saturating $p^{(B)}(k,l) \to 1$ across wide frequency bands during the transitional window.

2. **Recursive Feedback Delay:** Because $\tilde{\alpha}_d(k,l) \to 1$ whenever $p^{(B)}(k,l) \to 1$, the noise PSD update rate $(1-\tilde{\alpha}_d(k,l))$ approaches zero precisely when the step transition occurs. Full convergence to the new 0 dB noise level is therefore delayed until the $V$-frame buffer flushes out pre-transition history and the recursive smoother catches up.

### 5.3 Decoupling Amplitude Jumps via Scale-Invariant Learned SPP

System C resolves IMCRA's latency bottleneck by reformulating SPP estimation as an instantaneous pattern recognition task driven by scale-invariant spectro-temporal features. Rather than comparing current bin power against historical temporal minima, the Multi-Frame Contextual Neural Network evaluates relative spectral geometry within a short 3-frame causal receptive field $(t, t-1, t-2)$:

- **Invariance to Step-Function Noise Floor Shifts:** When the stationary fan noise power surges by +10 dB at $t = 3.0$ s, all frequency bins in a noise-only frame experience an approximately uniform additive shift in the log-power domain ($\Delta \log |Y(k,l)|^2 \approx +10$ dB). Because feature $f_1(k,l)$ subtracts the intra-frame median $\mathrm{median}_k(\log |Y(k,l)|^2)$, and features $f_2(k,l), f_3(k,l)$ subtract 7-bin and 23-bin local spectral averages, this global offset cancels out mathematically at the exact frame of transition ($l = l_{\text{trans}}$).

- **Harmonic Selectivity:** Speech energy concentrates in narrowband harmonic ridges and formants that exhibit high positive local spectral contrast ($f_2, f_3 \gg 0$), whereas broadband fan noise exhibits a flat local contrast ($f_2, f_3 \approx 0$) regardless of whether the SNR is 10 dB or 0 dB. Consequently, System C maintains a stable non-speech mean SPP baseline ($\sim$ 0.35–0.40) across the $t = 3.0$ s boundary without any upward step drift, enabling

rapid noise tracking ($\alpha_d + (1-\alpha_d)\hat{p}^{(C)}(k,l)^{0.8}$) and delivering a +3.6312 dB Post-3 s SNR Improvement—surpassing System B by +0.8183 dB and System A by +1.0144 dB.

### 5.4 Limitations and Engineering Trade-offs

Despite its superior objective performance, System C exhibits two practical limitations that warrant critical examination:

1. **Attenuation of Weak Unvoiced Consonants:** In the 4–8 kHz region of the time-frequency heatmap, unvoiced fricatives (such as /$s$/ and /$f$/) possess noise-like, non-harmonic spectra with low local spectral contrast. At 0 dB SNR, the MLP occasionally assigns lower SPP values ($\hat{p}^{(C)}(k,l) < 0.3$) to weak high-frequency fricative tails, causing mild over-suppression compared to voiced vowel formants below 4 kHz.

2. **Domain Generalization vs. Model Capacity:** While the compact 2-layer MLP (24 hidden units, 193 trainable parameters) guarantees zero-toolbox reproducibility, sub-second training, and negligible inference overhead, its capacity is tailored to broadband stationary/step-non-stationary noise profiles. In highly colored or impulsive noise environments (e.g., babble or keyboard clicks) that exhibit transient narrowband peaks, local spectral contrast alone may produce false-positive SPP spikes unless trained on broader multi-noise corpora.

## 6 Conclusion and Future Work

This study investigated the vulnerability of single-channel Wiener speech enhancement systems to abrupt non-stationary noise transitions and established a progressive three-stage solution framework. Through controlled experiments on a 10 dB $\rightarrow$ 0 dB step-transition dataset at $t = 3.0$ s, we demonstrated that:

1. Classical Hard-VAD Wiener filtering (**System A**) suffers from permanent threshold lock-up (VAD $\equiv 1$) after an upward noise step, freezing the noise PSD estimate and restricting Post-3 s SNR Improvement to +2.6168 dB (Global MSE: $0.9004 \times 10^{-3}$).

2. Statistical soft-decision tracking via IMCRA (**System B**) prevents catastrophic lock-up by updating noise PSD across inter-harmonic TF bins, achieving a Global Output SNR of 4.1983 dB (Global MSE: $0.8276 \times 10^{-3}$). However, its 0.8 s sliding-window minimum search introduces a transitional tracking lag that caps Post-3 s SNR Improvement at +2.8129 dB.

3. The proposed Lightweight Multi-Frame Contextual Neural Network (**System C**) eliminates minimum-tracking latency by predicting soft SPP from scale-invariant spectral contrast features. System C achieves the best overall performance across all metrics, reducing Global MSE to $\mathbf{0.7955 \times 10^{-3}}$, raising Global Output SNR to **4.3700** dB, and achieving a breakthrough Post-3 s SNR Improvement of $+\mathbf{3.6312}$ dB in the severe 0 dB noise region.

Future work will extend this hybrid DSP-neural paradigm along three directions: (i) integrating pitch-synchronous comb-filtering features and longer temporal dilated convolutions to preserve weak unvoiced consonants in the 4–8 kHz band; (ii) evaluating perceptual speech quality and intelligibility using standardized objective metrics including PESQ and STOI across multi-speaker datasets; and (iii) jointly optimizing the neural SPP estimator and the recursive Decision-Directed smoothing factor $\alpha_{dd}(k,l)$ in an end-to-end differentiable Wiener framework.

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